

Singular Relative Entropy Coding with Bits-Back Rejection Sampling

Gergely Flamich
Imperial College London
London, UK
g.flamich@imperial.ac.uk

Spencer Hill
Queen's University
Kingston, Canada
spencer.hill@queensu.ca

Abstract—A relative entropy code for a source $X \sim P_X$ is a stochastic code that encodes random samples from a prescribed $P_{Y|X}$ using as few bits as possible. A generalisation of entropy coding, it is a standard result that the minimum number of bits required to achieve this is at least the mutual information $I[X; Y]$. However, a particularly fascinating feature of relative entropy coding compared to entropy coding is that, in general, this lower bound is only achievable to within an additional logarithmic factor. As such, an important research direction is to identify channels where we can reduce this gap.

Sriramu and Wagner [1] achieved such success by exhibiting a relative entropy code for so-called singular channels with sub-logarithmic asymptotic redundancy. However, their code is quite involved and, sadly, cannot be implemented in practice.

In this paper, we construct the bits-back rejection sampler (BBRS), a relative entropy code that combines ideas from bits-back coding and (greedy) rejection sampling. Our analysis of BBRS reveals that the algorithm achieves the same asymptotic efficiency as Sriramu and Wagner's sampler, but with much simpler analysis and better constants. Moreover, BBRS can be implemented using standard relative entropy coding methods.

Index Terms—relative entropy coding, channel simulation, singular channel, rejection sampling, greedy rejection sampling, bits-back coding.

I. INTRODUCTION

Given a pair of dependent random variables $X, Y \sim P_{X,Y}$ defining the channel $X \rightarrow Y$, the goal of channel simulation is to find a *stochastic code* [2] that, given an input $X \sim P_X$, encodes a single sample $Y \sim P_{Y|X}$ in finitely many bits. More formally, a stochastic code is a triplet $(Z, \text{enc}, \text{dec})$, where

- Z is a random variable independent of X shared between the sender and receiver, called the *common randomness*.
- $\text{enc} : \mathcal{X} \times \mathcal{Z} \rightarrow \{0, 1\}^*$ is the *encoder* mapping an input and a realisation of Z to a variable-length binary code.
- $\text{dec} : \{0, 1\}^* \times \mathcal{Z} \rightarrow \mathcal{Y}$ is the *decoder* mapping a variable-length binary string and a realisation of Z to the output space $\mathcal{Y}$.

The basic identity that an (exact) stochastic code must satisfy is that for all inputs $x \in \mathcal{X}$,

$$\text{dec}(\text{enc}(x, Z), Z) \sim P_{Y|X=x} \quad (1)$$

The authors acknowledge financial support from Imperial College London through an Imperial College Research Fellowship grant awarded to GF. SH acknowledges support from the Natural Sciences and Engineering Research Council of Canada.

It is a standard result that the average codelength of any stochastic code must obey the bound [3], [4]

$$I[X; Y] \leq \mathbb{H}[Y | Z] \leq \mathbb{E}[|\text{enc}(X, Z)|] \quad (2)$$

Beginning with the work of Harsha et al. [3], there has been a natural line of work to construct stochastic codes with cost as close as possible to the mutual information lower bound in Equation (2). A significant milestone along this path is the work of Li and El Gamal [5], who describe a general scheme for constructing a stochastic code with expected codelength

$$\mathbb{E}[|\text{enc}(X, Z)|] \leq I[X; Y] + \log(I[X; Y] + 1) + 4 \quad (3)$$

Sadly, we cannot do better than Equation (3) in general: there exist channels for which any stochastic code requires at least $I[X; Y] + \log(I[X; Y] + 1) - \mathcal{O}(1)$ bits [5], [6]. As such, following [2], [7], we call stochastic codes that achieve a rate of $I[X; Y] + \mathcal{O}(\log(I[X; Y]))$ *relative entropy codes*.

The necessity of the logarithmic gap between the lower and upper bounds prompts a natural follow-up question: are there any broad classes of channels for which we can either prove a tighter lower bound or construct a relative entropy code with better rate? This question is precisely the focus of the works of Sriramu and Wagner [1] and Flamich, Sriramu and Wagner [8], and indeed of the present paper. These papers consider the “asymptotic setting,” meaning that for a pair of dependent random variables $X, Y \sim P_{X,Y}$ and product channel $X^n \rightarrow Y^n$, they study the best achievable rate

$$R_n \stackrel{\text{def}}{=} \frac{1}{n} \inf_{(Z, \text{enc}, \text{dec})} \mathbb{E}[|\text{enc}(X^n, Z)|] \quad (4)$$

By the construction of the independent and identically distributed (iid) channel, we have that $I[X^n; Y^n] = nI[X; Y]$. Combining this fact with Equations (2) and (3), we have that $R_n \rightarrow I[X; Y]$ as $n \rightarrow \infty$ and that this is optimal. Thus, the papers refine the analysis and study the asymptotic *logarithmic redundancy*

$$R^{\log} \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{n(R_n - I[X; Y])}{\log n} \quad (5)$$

Consulting Equations (2) and (3) once again, we immediately see that $0 \leq R^{\log} \leq 1$. The main contribution of Sriramu and Wagner's work [1] is identifying a simple condition called *singularity* (see Section II for the definition) on the

channel $X \rightarrow Y$ which completely determines $R^{\log}$. To be precise, they construct a family of relative entropy codes $\{(Z_n^{SW}, \text{enc}_n^{SW}, \text{dec}_n^{SW})\}$ based on rejection sampling whose asymptotic logarithmic redundancy for any channel $X^n \rightarrow Y^n$ satisfies

$$R_{SW}^{\log} \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} \frac{\mathbb{E}[\|\text{enc}_n^{SW}(X^n, Z_n)\|] - nI[X; Y]}{\log n} \leq \frac{1}{2} \quad (6)$$

Furthermore, when the channel $X \rightarrow Y$ is singular, they show that $0 \leq R^{\log} \leq R_{SW}^{\log} = 0$. This is complemented by further analysis in [1], [8], which shows that when $X \rightarrow Y$ is non-singular, we have the lower bound $\frac{1}{2} \leq R^{\log}$. Combining these results, we have the following clean characterisation of asymptotic logarithmic redundancy [1], [8]:

$$R^{\log} = \begin{cases} 0 & \text{when } X \rightarrow Y \text{ singular} \\ \frac{1}{2} & \text{when } X \rightarrow Y \text{ non-singular.} \end{cases} \quad (7)$$

However, the Sriramu-Wagner code for the achievability of Equation (7) has three clear shortcomings:

- 1) It relies on access to quantities that, despite being theoretically computable, are virtually impossible to compute, even for toy problems. This intractability immediately precludes any practical utility of the code.
- 2) Adding to the impracticality, the construction exhibits a great deal of “one-shot” inefficiency: its rate has several sublogarithmic terms and large constants that only vanish in the limit $n \rightarrow \infty$.
- 3) The reason *why* singular channels achieve zero asymptotic logarithmic redundancy is not very clear. Essentially, singularity causes certain terms to cancel in the rate analysis of the code, but there is no clear interpretation of why one ought to expect this to occur.

This paper addresses these shortcomings by constructing a stochastic code, which we call the *bits-back rejection sampler* (BBRS), and by providing a new proof of the achievability part of Equation (7) in the singular case. As we shall see, singularity plays a clear, central role in the construction of BBRS. Furthermore, although BBRS relies on the somewhat more advanced greedy rejection sampler, its one-shot rate is much better than the rate of the Sriramu-Wagner code. Finally, the quantities required to implement BBRS are easier to compute in practice, facilitating implementation.

A. Notation

In this paper, $\log$ denotes the binary logarithm. We denote the set of finite-length binary strings as $\{0, 1\}^*$, and for a string $s \in \{0, 1\}^*$ we denote its length as $|s|$. Capital Roman letters denote random variables and probability measures. Unless stated otherwise, we assume that all random variables are over some arbitrary Polish space. For two random variables X, Y and a probability measure P , we use a slight abuse of notation: $X \sim Y$ shall mean that X and Y are equal in distribution, while $X \sim P$ denotes that X has probability distribution P . For two probability measures Q, P where $Q \ll P$, dQ/dP denotes the Radon-Nikodym derivative of Q with respect to P . Bernoulli distributions with bias/success probability $p \in [0, 1]$

are denoted by $\text{Bern}(p)$, while geometric distributions with mean M are denoted by $\text{Geom}(M)$. The Shannon entropy of a discrete random variable is denoted as $\mathbb{H}[X]$ and the mutual information between two variables X, Y is denoted by $I[X; Y]$.

II. BACKGROUND

The five key concepts we require for our construction are *singular channels*, *rejection sampling*, *greedy rejection sampling*, *invertible sampling* and *bits-back coding*; we shall now cover them in this order.

A. Singular Channels

The basic motivation for singular channels is the observation that there exist certain channels for which all inputs that lead to a given output do so with the same probability. The generalisation for arbitrary $P_{Y|X}$ is the condition that only the support of $Y | X$ is affected by X , but not the precise likelihood:

Definition 1 (Singular channel [1], [9]). Let $X, Y \sim P_{X,Y}$ be a pair of dependent random variables. Then, we say that the channel $X \rightarrow Y$ is singular if there exists a P_Y -measurable function g such that

$$\frac{dP_{Y|X}}{dP_Y}(y | x) = g(y) \quad P_{X,Y}\text{-almost surely.} \quad (8)$$

The two canonical examples of singular channels are the additive uniform channel $Y = X + U$, where $U \sim \text{Unif}(-1, 1)$, and the binary erasure channel. Importantly, for the rest of this paper, we shall assume that the function g is known and shared by the communicating parties; this will be crucial for our construction of BBRS.

B. Rejection Sampling

Here we give a brief description of rejection sampling and how we may use it to construct a stochastic code; see [2] for a tutorial and [4] for a comprehensive survey of these concepts.

Originating in computational statistics, rejection sampling aims to simulate a sample from some prescribed target distribution Q given access to iid samples from some proposal distribution P . Concretely, let $r = dQ/dP$ and assume that r is bounded, that is, there exists $M \geq 1$ such that $\|r\|_\infty \leq M$. Then, at each step $k = 1, 2, \dots$, rejection sampling considers a sample $Z_k \sim P$, computes the *acceptance probability* $\alpha(Z_k) = r(Z_k)/M$, and flips a coin $B_k \sim \text{Bern}(\alpha(Z_k))$. If $B_k = 1$, the algorithm *accepts* Z_k and terminates, while if $B_k = 0$, it *rejects* Z_k and moves on to step $k + 1$. Letting K be the step in which the algorithm terminates, it is a standard result that $Z_K \sim Q$.

Given a pair of dependent random variables $X, Y \sim P_{X,Y}$, we can use rejection sampling to construct a stochastic code for the channel $X \rightarrow Y$: given an input $x \sim P_X$, we use rejection sampling to simulate a sample from the target $P_{Y|X}$ using P_Y as the proposal distribution. First, we set the common randomness as an iid sequence of P_Y -distributed proposal samples: $Z \leftarrow \{Z_i\}_{i=1}^\infty$. Then, letting $r_x(y) = \frac{dP_{Y|X}}{dP_Y}(y | x)$

Algorithm 1: Greedy rejection sampling, as formulated in [10]. See Section II for notation.

Input : Target Q , Shared iid samples $X_1, X_2 \dots \sim P$.
 $L_1, S_1 \leftarrow (0, 1)$
for $k = 1$ **to** ∞ **do**
 $\alpha_k \leftarrow \min \left\{ 1, \max \left\{ 0, \left(\frac{dQ}{dP}(X_k) - L_k \right) / S_k \right\} \right\}$
 $B_k \sim \text{Bern}(\alpha_k)$ **if** $B_k = 1$ **then**
 return X_k, k
 end
 $L_{k+1} \leftarrow L_k + S_k$
 $\mathcal{H}_{k+1} \leftarrow \left\{ y \in \Omega \mid L_{k+1} \leq \frac{dQ}{dP}(y) \right\}$
 $S_{k+1} \leftarrow Q(\mathcal{H}_{k+1}) - L_{k+1} \cdot P(\mathcal{H}_{k+1})$
end

with upper bound M_x , given some input x and the common randomness Z , we run rejection sampling to compute

$$K(x, Z) = \min\{k \in \mathbb{N} \mid B_k = 1, B_k \sim \text{Bern}(\alpha(x, Z_k))\} \quad (9)$$

with the acceptance probability $\alpha(x, z) = r_x(z)/M_x$. Given that each proposed sample Z_k and acceptance decision B_k is iid, we find that $K(x, Z) \sim \text{Geom}(M_x)$, that is, the acceptance step is geometrically distributed with mean M_x . Thus, assuming that both the sender and the receiver know M_x (an unrealistic assumption in general, but as we shall see, for BBRS it will not be an issue), we can construct an entropy code C (for example, a Huffman code) for $K(x, Z)$ whose average codelength is at most

$$\mathbb{E}[|C(K(x, Z))|] \leq \mathbb{H}[K(x, Z) \mid M_x] + 1 \leq \log M_x + 2 \quad (10)$$

Hence, our final rejection code for $X \rightarrow Y$ is

$$\begin{aligned} Z^{RS} &= \{Z_i\}, \quad Z_i \sim P_Y \\ \text{enc}^{RS}(x, Z^{RS}) &= C(K(x, Z^{RS})) \\ \text{dec}^{RS}(s, Z^{RS}) &= Z_{C^{-1}(s)} \quad Z_i \in Z^{RS} \end{aligned} \quad (11)$$

The issue with the stochastic code in Equation (11) is twofold: first, the required upper bound M_x is usually unknown to the decoder. On the other hand, if we use a universal upper bound $M \geq \sup_x M_x$, the rate will be much larger than the optimum. This issue is precisely why Harsha et al. developed greedy rejection sampling, which we shall discuss next.

C. Greedy Rejection Sampling (GRS)

Intuitively, the reason for the inefficiency of the rate of the rejection code in Equation (11) is that the acceptance decisions of rejection sampling are not only independent but also identically distributed. The key idea behind the greedy rejection sampler [3], [10] is to choose the acceptance probabilities according to the following criterion: in each step k , assuming the sampler hasn't terminated in a previous step, choose the acceptance probability $\alpha_k(z)$ to maximise the probability that the sampler terminates in the current step. Indeed, this design principle of greedily maximising the termination probability is the algorithm's namesake.

We describe GRS in Algorithm 1 for completeness; for a detailed discussion and analysis of the algorithm, see [3], [10]. For this paper, there are two important facts the reader needs to know. First, GRS is also a rejection procedure and, in essence, differs from regular rejection sampling only in that the acceptance probabilities are not identically distributed. Therefore, we can similarly use GRS to construct a stochastic code for a channel $X \rightarrow Y$. Second, letting

$$K'(x, Z) = \min\{k \in \mathbb{N} \mid B_k = 1, B_k \sim \text{Bern}(\alpha_k(x, Z_k))\}$$

as Theorem III.2 and in particular Equation 3 of [10] shows, we can construct an entropy code C' for K' such that

$$\mathbb{E}[|C'(K'(X, Z))|] \leq I[X; Y] + \log(I[X; Y] + 1) + 5 \quad (12)$$

As such, GRS is a general way of constructing a relative entropy code.

D. Invertible Sampling

The next ingredient we will need for our construction is invertible sampling, which sits at the heart of bits-back coding [10]–[12]. For this, we need to explicitly consider the bit stream s that the sender intends to transmit to the receiver and, accordingly, extend the definition of our codes. Concretely, henceforth we shall assume that for any discrete source distribution P we can construct a stream code $C_P : \{0, 1\}^* \times \mathcal{X} \rightarrow \{0, 1\}^*$ such that for any message stream $s \in \{0, 1\}^*$ and source $x \in \mathcal{X}$, we have

$$C_P^{-1}(C_P(s, x)) = (s, x) \quad (13)$$

and

$$|C_P(s, x)| - |s| \approx -\log P(x) \quad (14)$$

Examples of such stream codes are arithmetic coding [13] and asymmetric numeral systems (ANS) [14], [15].

The main idea behind invertible sampling is the answer to the following puzzle. Consider a very long string S whose bits are each iid $\text{Bern}(1/2)$ and a discrete distribution P . If we decode $(X, S') \leftarrow C_P^{-1}(S)$, what is the distribution of X ? To find the intuitive answer, consider the reverse scenario first: encoding some $X' \sim P$ into the stream of uniform bits S' . As we assumed that C_P is a stream code and we are encoding a source sample X' using its own distribution into S' , $C_P(S', X')$ ought to be a string of uniformly random bits. Thus, if we now invert the procedure and start with S , and put $(X, S') = C_P^{-1}(S)$, then we ought to have $X \sim P$. These arguments can be made precise, for example, see [16, Chapter 13.3], [15] and [17].

The reason why we call this notion *invertible sampling* is because starting from a pseudo-random number generation (PRNG) perspective, we may consider the initial stream S as the random seed or random state of a PRNG algorithm and the decoding procedure $(X, S') \leftarrow C_P^{-1}(S)$ as simulating a random sample and advancing the random seed/state of the algorithm. However, we can do more than usual: we can invert the sampler and recover the original seed S by re-encoding X into the stream S' using C_P !

E. Bits-Back Coding

The purpose of bits-back coding is to construct a stream code for some source $X \sim P_X$ where P_X is difficult to compute, but there exists a channel $P_{Y|X}$ and function g such that it is easy to perturb X using $X \rightarrow Y$ and recover X by computing $g(Y)$. Indeed, a high-level description of bits-back coding is [7]

invertible sampling + error correction = bits-back coding.

More concretely, as a toy example, consider X to be uniformly distributed over the codewords of the (7, 4, 3)-Hamming code and let $Y = X + \epsilon$, where ϵ is uniformly distributed over the Hamming ball of radius 1 over 7-bit strings. Observe that we constructed this channel such that we can always recover X from Y by the very definition of Hamming codes; indeed, g is the Hamming decoder. Furthermore, we see that the marginal P_Y is the uniform distribution over 7-bit strings.

Consider now constructing an entropy code for X . Of course, in this toy setting, we could do this directly, but note that we can also take advantage of the error-correction procedure combined with invertible sampling as follows:

- 1) Assume the encoder has stream S and receives a new symbol $X \sim P_X$ to encode.
- 2) They **decode** $(S', Y) \leftarrow C_{P_{Y|X}}^{-1}(S)$. Then, $Y \sim P_{Y|X}$.
- 3) They **encode** $S'' \leftarrow C_{P_Y}(S', Y)$ and send S'' .

Somewhat magically, this procedure is completely invertible:

- 1) Upon receiving S'' , the decoder uses the shared coding distribution P_Y to recover $(S', Y) \leftarrow C_{P_Y}^{-1}(S'')$
- 2) They “error-correct” Y by computing $X \leftarrow g(Y)$.
- 3) Finally, they recover the original stream by re-encoding Y into S' : they compute $S \leftarrow C_{P_{Y|X}}(S', Y)$, which they can do, since they now have access to $P_{Y|X}$.

Note that at the end of the decoding procedure, the decoder has obtained the source X as well as the original stream state S . Furthermore, the encoding procedure is also efficient: decoding (S', Y) from the stream S decreases the length of the stream by $\approx -\log P_{Y|X}(Y | X)$ bits, while encoding Y into S' using P_Y increases the stream length by $\approx -\log P_Y(Y)$ bits. Thus, in total, the stream length increases by roughly

$$-\log \frac{P_Y(Y)}{P_{Y|X}(Y | X)} = -\log \frac{P_X(X)}{P_{X|Y}(X | Y)} = -\log P_X(X)$$

bits, where the first equality follows from Bayes’ rule and the second follows from the fact that X is a deterministic function of Y and hence the denominator is simply equal to 1.

The most widespread application of bits-back coding is in the construction of the ANS stream code [15], but it is also used in shuffle coding, a state-of-the-art entropy code for unordered/exchangeable data [18], and bits-back quantisation, a relative entropy code for truncated distributions over the reals [10]. Finally, we note that bits-back coding can also be applied when X cannot be fully recovered from Y : assuming we can only recover $\hat{X} = g(Y)$, we can then encode $X | \hat{X}$. However, we note that in this case, the procedure is no longer rate-optimal. Indeed, this was the setting in which bits-back

coding was first introduced to practical machine learning-based lossless compression by Townsend et al. [12].

III. BITS-BACK REJECTION SAMPLING

We are now finally ready to present our stochastic code for singular channels, bits-back rejection sampling (BBRS).

The high-level idea is very similar to that of the Sriramu-Wagner construction: instead of encoding a sample $Y \sim P_{Y|X}$ directly, we produce a two-part code by first encoding the quantised log-density ratio $\Gamma = Q_\Delta \left(\log \frac{dP_{Y|X}}{dP_Y}(Y | X) \right)$ to Δ precision, and then encode a sample $Y | X, \Gamma$. As elucidated in [1], the second sample can be communicated optimally using standard rejection sampling, as the density ratio Γ , and therefore the acceptance probability, is fixed and known to both encoder and decoder.

However, our construction is based on the observation that when $X \rightarrow Y$ is singular, we can use the measurable function g from Definition 1 as an “error-correction mechanism” to recover Γ from Y . Therefore, we can apply bits-back coding to recover almost all the bits we used to encode Γ !

Concretely, let $P_{X,Y}$ be given with $X \rightarrow Y$ singular. For $c \in \mathbb{R}$ and $\Delta > 0$, let $Q_\Delta(c) = \Delta \cdot \lfloor c/\Delta \rfloor$ be a Δ -fine quantiser for the reals. Let $\Gamma = Q_\Delta \left(\log \frac{dP_{Y|X}}{dP_Y}(Y | X) \right)$. Then, the encoding procedure is:

- 1) Given $X \sim P_X$, the encoder simulates $\Gamma \sim P_{\Gamma|X}$.
- 2) Next, they use a GRS-based relative entropy code with P_Y -distributed shared randomness $\{Z_k\}_{k=1}^\infty$ to encode a sample $Y' \sim P_{Y|\Gamma}$. Concretely, they encode GRS’s returned index K using an appropriate entropy code.
- 3) The encoder knows that given Y' , the singularity of $X \rightarrow Y$ allows the decoder to recover Γ by computing $\Gamma = Q_\Delta(\log g(Y'))$. Thus, they use a simple rejection code to encode a sample from $Y | X, \Gamma$ using $P_{Y|\Gamma}$ -distributed common randomness $\{\Upsilon_n\}_{n=1}^\infty$. To implement the rejection code, they compute two quantities:

$$M(\Gamma) = 2^{\Gamma+\Delta} \quad (15)$$

$$M(\Gamma, X) = M(\Gamma) / \frac{dP_{X|\Gamma}}{dP_X}(X | \Gamma) \quad (16)$$

It is not hard to show that $M(\Gamma, X)$ is an upper bound to $\frac{dP_{Y|X,\Gamma}}{dP_{Y|\Gamma}}$ and can therefore be used to run the rejection sampler [1]; we provide a proof of this fact in Appendix A for completeness. Finally, they encode the rejection sampler’s returned index N using an entropy code for the geometric distribution with mean $M(\Gamma)$.

The encoder also knows that once the decoder recovers Γ , they can “replay” the GRS procedure for sampling $Y | \Gamma$. This observation then motivates a bits-back procedure: given the encoder’s message stream s , the encoder **decodes the GRS acceptance decisions** from s , thus shortening the message. They can do this because they know the decoder can re-encode the acceptance decisions back into the stream once it has recovered Γ . Although the above is a full description of BBRS, for clarity, we also provide detailed pseudocode for the encoder and decoder in Algorithms 2 and 3, respectively. We

use the ANS stream code [14], [15] as our invertible sampler in the pseudocode, which is a last-in-first-out code: the last symbol we encode into the stream is the first symbol we decode from it. Note that Algorithms 2 and 3 account for this by encoding the two-part code in the appropriate order.

A. Rate Analysis of Bits-Back Rejection Sampling

We now turn to the analysis of the BBRs rate and obtain

Theorem 1 (One-shot rate of BBRs). *Let $X, Y \sim P_{X,Y}$ be a pair of dependent random variables with $X \rightarrow Y$ singular. Let $(Z, \text{enc}, \text{dec})$ be the stochastic code defined by Algorithms 2 and 3. Then, we have*

$$\mathbb{E}[\|\text{enc}(X, Z)\|] \leq I[X; Y] + \log(\mathbb{H}[\Gamma] + 1) + 2\Delta + 5 \quad (17)$$

Proof. The rate of the algorithm consists of three parts:

- 1) The number of bits required to encode the GRS index K , which encodes the sample $Y' \sim P_{Y|\Gamma}$.
- 2) The index N which encodes $Y \sim P_{Y|X,\Gamma}$.
- 3) The number of bits by which the message is reduced from decoding the GRS acceptance decisions.

Thus, we now analyse these quantities in this order.

Rate required for K . From Section II-C, and Equation (12) in particular, assuming the sender and receiver both know $I[Y; \Gamma]$, the number of bits required to encode K is at most $I[Y; \Gamma] + \log(I[Y; \Gamma] + 1) + 4 = \mathbb{H}[\Gamma] + \log(\mathbb{H}[\Gamma] + 1) + 4$

where the equality holds as Γ is a function of Y .

Rate required for N . As we explained in Section II-B, since we use standard rejection sampling with upper bound $M(X, \Gamma)$, N will be geometrically distributed with mean $M(X, \Gamma)$. Recall that the sender encodes N using a geometric random variable with mean $M(\Gamma)$. Then, noting that $\mathbb{E}[M(\Gamma, X) | \Gamma] = M(\Gamma)$, the average cost of encoding N will be equal to the expected cross entropy

$$\begin{aligned} & \mathbb{E}_X[\mathbb{H}[\text{Geom}(M(X, \Gamma)) | \text{Geom}(M(\Gamma))]] \\ &= \mathbb{E}_{X,\Gamma}[\log M(\Gamma) - (M(X, \Gamma) - 1) \log(1 - 1/M(\Gamma))] \\ &= \mathbb{E}_\Gamma[\log M(\Gamma) - (M(\Gamma) - 1) \log(1 - 1/M(\Gamma))] \\ &= \mathbb{H}[\text{Geom}(M(\Gamma))] \\ &\leq \mathbb{E}[\log M(\Gamma)] + 1 \\ &= \mathbb{E}[\Gamma + \Delta] + 1 \\ &\leq I[X; Y] + 2\Delta + 1 \end{aligned}$$

Number of bits decoded to sample $Y' \sim P_{Y|\Gamma}$. Letting A_k denote the acceptance probabilities of GRC, for a fixed $K = k$ the number of decoded bits H is

$$H = -\log A_k - \sum_{i=1}^{k-1} \log(1 - A_i) = -\log \left(A_k \prod_{i=1}^{k-1} (1 - A_i) \right)$$

However, note that $A_k \prod_{i=1}^{k-1} (1 - A_i) = \mathbb{P}[K = k | \{Z_k\}_{k=1}^\infty]$. Hence, $\mathbb{E}[H] = \mathbb{H}[K | \{Z_k\}_{k=1}^\infty] \geq I[Y; \Gamma] = \mathbb{H}[\Gamma]$, where the inequality follows from Equation (2) and the last equality follows since Γ is a deterministic function of Y .

Putting it all together. Summing the upper bounds on the rates for K and N and subtracting the lower bound on the number of decoded bits yields the desired result. $\square$

To recover Sriramu and Wagner's upper bound, we use their Lemma 5 from [1], which essentially follows directly from the central limit theorem applied to $\log \frac{dP_{Y^n|X^n}}{dP_Y^n}(Y^n | X^n)$:

Lemma 2. *Let $\Gamma_n = Q_\Delta \left(\log \frac{dP_{Y^n|X^n}}{dP_Y^n}(Y^n | X^n) \right)$. Then, there is a constant $C > 0$ that depends on $P_{X,Y}$ and Δ , but not n , such that*

$$\mathbb{H}[\Gamma_n] \leq \frac{1}{2} \log n + C \quad (18)$$

Thus, combining Theorem 1 with Lemma 2, we immediately obtain the desired corollary.

Corollary 3. *Let $X, Y \sim P_{X,Y}$ be a pair of dependent random variables with $X \rightarrow Y$ singular. Consider the family of stochastic codes $\{(Z_n, \text{enc}_n, \text{dec}_n)\}_{n=1}^\infty$ with $(Z_n, \text{enc}_n, \text{dec}_n)$ the stochastic code formed by applying Algorithms 2 and 3 sequentially to encode n copies of $X \rightarrow Y$. Then, the logarithmic redundancy of this family is*

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[\|\text{enc}_n(X^n, Z_n)\|] - nI[X; Y]}{\log n} = 0 \quad (19)$$

which also implies that the optimal logarithmic redundancy is $R^{\log} = 0$ as well.

Finally, we note that BBRs can be extended to non-singular channels as well by encoding $\Gamma | Y$; this is akin to the “encoding the residuals” idea we discussed in Section II-E. Sadly, this scheme is no longer efficient, as it does not even achieve the 1/2-logarithmic redundancy asymptotically, and thus we do not discuss it further.

However, note that our main motivation for developing BBRs was to give a simpler code than the Sriramu-Wagner construction, but the non-singular case already has simple algorithms that achieve the optimum. As Sriramu and Wagner point out in their paper, in an observation attributed to Lucas Theis [1], the 1/2-logarithmic redundancy can also be achieved by a small modification to existing relative entropy codes, such as GRS or the Poisson functional representation (PFR). However, as they do not provide an explicit construction, we provide one for PFR in Appendix B for completeness.

IV. CONCLUSION

We constructed a stochastic code for singular channels using a sampling algorithm we call the bits-back rejection sampler. We analysed its rate and showed that in the asymptotic case, its logarithmic redundancy is zero, thereby recovering Sriramu and Wagner's result for singular channels. However, our sampler uses singularity more explicitly than their code by embedding it in a bits-back procedure.

One unsatisfactory aspect of our sampler, which it shares with Sriramu and Wagner's construction, is that it relies on some arbitrary discretisation of the log-density ratio. As such, future work might aim either to eliminate this quantisation or to analyse its precise role in the sampler's behaviour.

REFERENCES

- [1] S. M. Sriramu and A. B. Wagner, “Optimal redundancy in exact channel synthesis,” in *2024 International Symposium on Information Theory (ISIT)*, pp. 1913–1918, IEEE, 2024.
- [2] G. Flamich and D. Gündüz, “Data compression with stochastic codes,” *arXiv preprint arXiv:2602.07635*, 2026.
- [3] P. Harsha, R. Jain, D. McAllester, and J. Radhakrishnan, “The communication complexity of correlation,” *IEEE Transactions on Information Theory*, vol. 56, no. 1, pp. 438–449, 2010.
- [4] C. T. Li, “Channel simulation: Theory and applications to lossy compression and differential privacy,” *Foundations and Trends® in Communications and Information Theory*, vol. 21, no. 6, pp. 847–1106, 2024.
- [5] C. T. Li and A. El Gamal, “Strong functional representation lemma and applications to coding theorems,” *IEEE Transactions on Information Theory*, vol. 64, no. 11, pp. 6967–6978, 2018.
- [6] M. Braverman and A. Garg, “Public vs private coin in bounded-round information,” in *International Colloquium on Automata, Languages, and Programming*, pp. 502–513, Springer, 2014.
- [7] G. Flamich, *Data Compression with Relative Entropy Coding*. PhD thesis, University of Cambridge (United Kingdom), 2024.
- [8] G. Flamich, S. M. Sriramu, and A. B. Wagner, “The redundancy of non-singular channel simulation,” in *2025 IEEE International Symposium on Information Theory (ISIT)*, pp. 1–6, IEEE, 2025.
- [9] Y. Altuğ and A. B. Wagner, “Refinement of the random coding bound,” *IEEE Transactions on Information Theory*, vol. 60, no. 10, pp. 6005–6023, 2014.
- [10] G. Flamich and L. Theis, “Adaptive greedy rejection sampling,” in *2023 IEEE International Symposium on Information Theory (ISIT)*, pp. 454–459, IEEE, 2023.
- [11] G. E. Hinton and D. Van Camp, “Keeping the neural networks simple by minimizing the description length of the weights,” in *Proceedings of the sixth annual conference on Computational learning theory*, pp. 5–13, 1993.
- [12] J. Townsend, T. Bird, and D. Barber, “Practical lossless compression with latent variables using bits back coding,” in *International Conference on Learning Representations*, 2019.
- [13] I. H. Witten, R. M. Neal, and J. G. Cleary, “Arithmetic coding for data compression,” *Communications of the ACM*, vol. 30, no. 6, pp. 520–540, 1987.
- [14] J. Duda, K. Tahboub, N. J. Gadgil, and E. J. Delp, “The use of asymmetric numeral systems as an accurate replacement for huffman coding,” in *2015 Picture Coding Symposium (PCS)*, pp. 65–69, IEEE, 2015.
- [15] R. Bamler, “Understanding entropy coding with asymmetric numeral systems (ans): a statistician’s perspective,” *arXiv preprint arXiv:2201.01741*, 2022.
- [16] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. John Wiley & Sons, 2012.
- [17] D. Knuth, “The complexity of nonuniform random number generation,” *Algorithm and Complexity, New Directions and Results*, 1976.
- [18] J. Kunze, D. Severo, G. Zani, J.-W. van de Meent, and J. Townsend, “Entropy coding of unordered data structures,” in *The Twelfth International Conference on Learning Representations*, 2024.
- [19] C. T. Li and V. Anantharam, “A unified framework for one-shot achievability via the poisson matching lemma,” *IEEE Transactions on Information Theory*, vol. 67, no. 5, pp. 2624–2651, 2021.

APPENDIX A
MINOR TECHNICAL DETAILS FOR BBRs

Here, we show that $M(X, \Gamma)$ is a valid upper bound on the density ratio of rejection sampling in the second stage. This verifies the correctness of the scheme. Our proof follows closely that of [1, Lemma 3]. We rearrange the Radon-Nikodym derivative:

$$\begin{aligned} \frac{dP_{Y|X, \Gamma}}{dP_{Y|\Gamma}}(y | x, \gamma) &= \frac{dP_{\Gamma|X, Y}}{dP_{\Gamma|Y}}(\gamma | x, y) \frac{dP_{Y|X}}{dP_Y}(y | x) \frac{dP_{\Gamma}}{dP_{\Gamma|X}}(\gamma | x) \quad (20) \\ &= \frac{P_{\Gamma|X, Y}(\gamma | x, y)}{P_{\Gamma|Y}(\gamma | y)} \frac{dP_{Y|X}}{dP_Y}(y | x) \frac{P_{\Gamma}(\gamma)}{P_{\Gamma|X}(\gamma | x)} \quad (21) \\ &= \frac{P_{\Gamma}(\gamma)}{P_{\Gamma|X}(\gamma | x)} \frac{dP_{Y|X}}{dP_Y}(y | x) \quad (22) \end{aligned}$$

In the above, Equation (20) follows by the chain rule for the Radon-Nikodym derivative; Equation (21) follows because Γ is discrete and therefore P_{Γ} , $P_{\Gamma|X}$, $P_{\Gamma|Y}$ and $P_{\Gamma|X, Y}$ are absolutely continuous with respect to the counting measure; and Equation (22) holds because $X \rightarrow Y$ is singular and hence $P_{\Gamma|X, Y}$ and $P_{\Gamma|Y}$ are delta functions. Then:

$$\begin{aligned} \sup_{y: \Gamma=y} \frac{dP_{Y|X, \Gamma}}{dP_{Y|\Gamma}}(y | x, \gamma) &\leq M(\Gamma) / \frac{dP_{X|\Gamma}}{P_X}(X | \Gamma) = M(\Gamma, X). \quad (23) \end{aligned}$$

APPENDIX B
IMPROVED POISSON FUNCTIONAL REPRESENTATION
CODING SCHEME

Here, we give an explicit construction that the Poisson functional representation (PFR) can be moderately modified so that its code length attains the desired $1/2$ -logarithmic redundancy. We do not describe PFR here; see the original paper [5] for a description or [2], [4], [7] for a more detailed account. The reader only needs to know that, akin to rejection sampling and GRS, PFR also uses P_Y -distributed common randomness $\{Z_k\}_{k=1}^{\infty}$ as proposal samples for the target distribution $P_{Y|X}$ and encodes the index K of the selected sample. Then, letting

$$R_x = \frac{dP_{Y|X}}{dP_Y}(Z_K | x) \quad (24)$$

$$\Gamma_x = Q_{\Delta}(\log R_x) \quad (25)$$

$$M(\Gamma_x) = 2^{\Gamma_x + \Delta} + 1 \quad (26)$$

$$M'(R_x) = \mathbb{E}_{Z \sim P_Y} \left[\max \left\{ \frac{dP_{Y|X}}{dP_Y}(Z | x), R_x \right\} \right] \quad (27)$$

we have the following result [4], [7], [19]:

$$K | R_x \sim \text{Geom}(M'(R_x)) \quad (28)$$

Thus, consider now encoding K using a geometric distribution with mean $M(\Gamma_x)$. Then, the rate will be equal to the expected cross-entropy

$$\begin{aligned} \mathbb{E}_X [\mathbb{H}[\text{Geom}(M'(R_X)) | \text{Geom}(M(\Gamma_X))]] \\ = \mathbb{E}[\log M(\Gamma_X) - (M'(R_X) - 1) \log(1 - 1/M(\Gamma_X))] \end{aligned}$$

Treating these terms separately, for the first term, we see that

$$\begin{aligned} \mathbb{E}[\log M(\Gamma_X)] &= \mathbb{E}[\log(2^{\Gamma_X + \Delta} + 1)] \\ &\leq \mathbb{E}[\log(R_X \cdot 2^{\Delta} + 1)] \\ &= \mathbb{E}[\log R_X] + \mathbb{E}[\log(2^{\Delta} + 1/R_X)] \\ &\leq \mathbb{E}[\log R_X] + \mathbb{E}[\log(2^{\Delta} e^{1/R_X})] \\ &= I[X; Y] + \Delta + \log e \cdot \mathbb{E}_{X, Y \sim P_{X, Y}} \left[\frac{dP_Y}{dP_{Y|X}}(Y | X) \right] \\ &= I[X; Y] + \Delta + \log e \end{aligned}$$

For the second term, we first note that

$$\begin{aligned} M'(R_x) &= \mathbb{E}_{Z \sim P_Y} \left[\max \left\{ \frac{dP_{Y|X}}{dP_Y}(Z | x), R_x \right\} \right] \quad (29) \\ &\leq R_x + 1 \end{aligned}$$

and also that $R_x + 1 \leq M(\Gamma_X)$, which implies

$$-\log \left(1 - \frac{1}{M(\Gamma_x)} \right) \leq -\log \left(1 - \frac{1}{R_x + 1} \right) \quad (30)$$

Applying Equations (29) and (30) to the second term, we see that it is bounded above as

$$\begin{aligned} -\mathbb{E}[(M'(R_X) - 1) \log(1 - 1/M(\Gamma_X))] \\ \leq -\mathbb{E} \left[((R_X + 1) - 1) \cdot \log \left(1 - \frac{1}{R_X + 1} \right) \right] \\ \leq \log e \end{aligned}$$

where the second inequality follows from the standard point-wise inequality: for $x \in (0, 1)$, we have $-x \ln x \leq 1$. Finally, this shows that the cross-entropy is bounded above by

$$\begin{aligned} \mathbb{H}[\text{Geom}(M'(R_X)) | \text{Geom}(M(\Gamma_X))] \\ \leq I[X; Y] + \Delta + 2 \log e \quad (31) \end{aligned}$$

Thus, we can construct a two-part code by first encoding Γ_X followed by $K | R_X$. The total rate will thus be less than

$$I[X; Y] + \mathbb{H}[\Gamma] + \Delta + 2 \log e + 1$$

From this, we see that in the asymptotic setting, the only quantity contributing to the logarithmic redundancy is $\mathbb{H}[\Gamma_n]$, which by Lemma 5 of Sriramu and Wagner grows as $\frac{1}{2} \log n + \mathcal{O}(1)$, as desired.

Algorithm 2: Bits-back rejection sampling encoder

Input:

Channel input $X = x$
Common randomness $\{Z_k\}$ with $Z_k \sim P_Y$ for each k
Collection of common randomness $\{\{\Upsilon_n\}_{\Gamma=\gamma}\}_\gamma$,
where for each sequence $\{\Upsilon_n\}_{\Gamma=\gamma}$, we have
 $\Upsilon_n \sim P_{Y|\Gamma=\gamma}$ for each n
Message bit-stream s

Output:

Message bit-stream s with two-part code K, N
embedded in it with $Z_K \sim P_{Y|\Gamma}$ and $\Upsilon_N \sim P_{Y|X,\Gamma}$

// Stage 1: Sample the quantised
density ratio

$\hat{Y} \sim P_{Y|X=x}$
 $\Gamma \leftarrow Q_\Delta \left(\log \frac{dP_{Y|X}}{dP_Y}(\hat{Y} | x) \right)$

// Stage 2: GRS to obtain an index K
such that $Z_K \sim P_{Y|\Gamma}$
 $k, b \leftarrow 1, 0$

while $b = 0$ **do**

// Compute GRS acceptance
probability. Note that S_k and
 L_k only depend on $\frac{dP_{Y|\Gamma}}{dP_Y}$

$A_k \leftarrow$
 $\min \left\{ 1, \max \left\{ 0, \left(\frac{dP_{Y|\Gamma}}{dP_Y}(Z_k) - L_k \right) / S_k \right\} \right\}$

// Obtain acceptance decision and
updated stream with invertible
sampler, shortening the stream

$s, B_k \leftarrow \text{ANSdecode}(s | \text{Bern}(A_k))$

$b \leftarrow B_k$

$k \leftarrow k + 1$

end

// Stage 3: rejection sampling an
index N such that $\Upsilon_N \sim P_{Y|X,\Gamma}$
 $M(\Gamma) \leftarrow 2^{\Gamma+\Delta}$

$M(\Gamma, X) \leftarrow M(\Gamma) / \frac{dP_{X|\Gamma}}{dP_X}(X | \Gamma)$

Select $\{\Upsilon_n\}_\Gamma$ from the collection of common
randomness, i.e. where $\Upsilon_n \sim P_{Y|\Gamma}$

for $n = 1, 2, \dots$ **do**

$U_n \sim \text{Unif}(0, 1)$

if $U_n \leq \frac{dP_{Y|\Gamma,X}}{dP_{Y|\Gamma}}(\Upsilon_n | \Gamma, x) / M(\Gamma, X)$:

// Upon acceptance, encode K, N
into the stream

$s \leftarrow \text{ANSencode}(s, N | \text{Geom}(M(\Gamma)))$

$s \leftarrow \text{ANSencode}(s, K | \zeta)$ // Use
optimal zeta code

return s

end**end**

Algorithm 3: Bits-back rejection sampling decoder

Input:

Message bit-stream s
Common randomness $\{Z_k\}$ with $Z_k \sim P_Y$ for each k
Collection of common randomness $\{\{\Upsilon_n\}_{\Gamma=\gamma}\}_\gamma$,
where for each sequence $\{\Upsilon_n\}_{\Gamma=\gamma}$, we have
 $\Upsilon_n \sim P_{Y|\Gamma=\gamma}$ for each n

Output:

$Y \sim P_{Y|\Gamma,X=x}$ and original message stream s

// Stage 1: recover K and hence

$Z_K \sim P_{Y|\Gamma}$

$(s, K) \leftarrow \text{ANSdecode}(s | \zeta)$

$\hat{Y} \leftarrow Z_K$

// Stage 2: compute Γ using the
singularity of the channel; see
Equation (8).

$\Gamma \leftarrow Q_\Delta(\log g(\hat{Y}))$

// Stage 3: recover N and hence

$\Upsilon_N \sim P_{Y|X,\Gamma}$

$M(\Gamma) \leftarrow 2^{\Gamma+\Delta}$

$N \leftarrow \text{ANSdecode}(s | \text{Geom}(M(\Gamma)))$

// Stage 4: re-encode the GRS
decision bits into the stream

for $k = K, K - 1, \dots, 1$ **do**

// Compute acceptance probability

$A_k \leftarrow$
 $\min \left\{ 1, \max \left\{ 0, \left(\frac{dP_{Y|\Gamma}}{dP_Y}(Z_k) - L_k \right) / S_k \right\} \right\}$

// We know the K th decision was an
accept, so re-encode a 1. All
decisions before the K th one
were rejects, so re-encode a 0.

$s \leftarrow \text{ANSencode}(s, 1 \text{ if } k = K \text{ else } 0 |$
 $\text{Bern}(A_k))$

end

// Υ_N is selected from the common
randomness $\{\Upsilon_n\}$ where $\Upsilon_n \sim P_{Y|\Gamma}$

return Υ_N, s
